IMAGE PROCESSING, COMPUTER VISION AND PATTERN RECOGNITION
(CT036-3-IPPR)
IN–COURSE ASSESSMENT
(ASSIGNMENT - GROUP ASSIGNMENT)

Intake	**:**	**APU3F/D3F/APU3F/D3F/2609CS/CS(AI)**
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Due date	**:**	**13th Dec, 2026**

IMAGE SEGMENTATION, DETECTION AND RECOGNITION

Automated Abdominal CT Organ & Lesion Diagnostics Pipeline

1.0 THE ASSIGNMENT OVERVIEW & objectives

In clinical abdominal radiology, Computed Tomography (CT) scans display a wide dynamic range of tissue densities recorded in Hounsfield Units (HU). Raw scans frequently suffer from low soft-tissue contrast, streaking artifacts, and complex background organ interference.

In this assignment, your group will engineer an end-to-end Computer-Aided Diagnosis (CAD) pipeline. The system must ingest raw abdominal CT slice data, apply radiometric windowing and 辐射窗技术 enhancement, isolate target abdominal organs (Liver or Kidneys) and potential lesions using classical segmentation, and classify pathological regions using spatial, morphological, and GLCM texture features. 病变区域 空间特征 形体特征 灰度共生矩阵纹理特征

三维CT图像》使用辐射窗技术来截取和放大目标区域》对图片进行降噪和提高对比度

2.0 OBJECTIVES OF THIS PROJECT:

Overall objectives: Equip students with practical knowledge about image processing, computer vision and pattern recognition in programming and computing. The focus will not be on teaching all of the details of programming under specific platforms, but rather on providing both a high-level understanding and practical implementation experience of reusable algorithms and coding techniques that apply to the development of applications across different platforms and genres.

Project specific objectives:

Provide a solution to automatically detect, segment, and classify (Kidney/Liver, and Tumor) through image processing and computer vision.

3.0 LEARNING OUTCOMES:

At the end of this project, the students should be able to:

- Design and implement a software artifact that processes images for computer vision and pattern recognition for a given problem domain using appropriate methods.
- Make a selection between candidate techniques based upon a rational critical evaluation of the requirements of a particular problem.
- **CT Radiometry & Enhancement:** Apply image enhancement techniques, intensity normalization, and spatial/frequency filtering to emphasize soft-tissue structures while suppressing CT reconstruction noise.
- **Abdominal Segmentation:** Implement and compare classical edge-, threshold-, morphological-, and region-based segmentation algorithms to isolate the liver or kidney boundaries and internal focal lesions.
- **Feature Extraction:** Compute shape descriptors and Grey-Level Co-occurrence Matrix (GLCM) texture metrics from segmented organ/lesion masks.
- **Pattern Recognition:** Build and evaluate a classical pattern classifier to categorize soft-tissue ROI states (e.g., *Healthy Tissue vs. Cyst/Tumor*).
- **Quantitative Validation:** Benchmark pipeline performance against ground-truth radiologist masks using standard overlapping and statistical metrics.

4.0 APPROVED PUBLIC DATASET

Your group must select one of the following benchmarks 2D CT datasets (or 2D axial slice extractions from 3D volumes):

Option A: LiTS (Liver Tumor Segmentation Benchmark)

- **Application:** Liver parenchyma segmentation and hepatic lesion detection.

- **Classes:** Background, Healthy Liver Tissue, Hepatic Lesion/Tumor.
- **Imaging Challenges:** Low contrast between liver tissue and hypo-dense lesions, overlapping intensity profiles with adjacent stomach/spleen tissue.
- https://github.com/Auggen21/LITS-Challenge

Option B: KiTS (Kidney Tumor Segmentation Benchmark)

- **Application:** Renal boundary extraction and solid/cystic tumor identification.
- **Classes:** Background, Healthy Kidney, Renal Cyst/Tumor.
- **Imaging Challenges:** Complex structural boundaries, streak artifacts near spine/ribs, variable lesion sizes.
- https://kits-challenge.org/kits23/

5.0 GENERAL GUIDELINES

This assignment is to be carried out as group work. Students will also be assessed individually on their contributions to the assignment.

- Each group must consist of 4 members. Groups should be formed within 2 weeks of the assignment starting date.
- The group must select one dataset (kidney (KiTs) or Liver (LiTs).
- Each member should work on minimum one organ (Kidney or tumor).
- Each member should use different algorithms in each phase; two members can't use the same algorithm.
- Each member should work on different images from the dataset (Different cases/patients).
- Testing dataset should be at least 10 images for each organ.
- The system must **not** be sensitive to environmental changes (**Haar Cascade, Tensorflow and pattern matching methods should not be used)**.
- Deep learning or pre-trained model **should NOT** be used. Only classical algorithms for segmentation.

6.0 ASSIGNMENT WORKFLOW & TECNHICAL REQUIREMENTS

[Raw CT Slice] ──►
[Phase 1: Image Enhancement] ──►
[Phase 2: Organ/Lesion Segmentation] ──►
[Phase 3: Pattern Recognition] ──► [Diagnostic Class]

Phase 1: CT Preprocessing & Spatial/Frequency Enhancement

Goal: Convert raw attenuation values into optimized soft-tissue grayscale representations and remove noise.

1. **Image enhancement & Intensity Normalization:**
    - Normalize the clipped array to an 8-bit grayscale dynamic range (0--255)
2. **Spatial Domain Filtering:**

    Each member should use different algorithms for different organs (Kidney/stone/tumor)

    - Compare edge-preserving filtering (Bilateral Filter or Anisotropic Diffusion) against standard Median or Gaussian smoothing to attenuate CT streaking noise without blurring organ edges.
    - Enhance the contrast using Histogram by applying Contrast Limited Adaptive Histogram Equalization ***(CLAHE)*** on the windowed soft-tissue band and ***Histogram Equalization*** and **compare the results**.
3. **Frequency Domain Enhancement:**
    - Compute the 2D Discrete Fourier Transform (2D-DFT) and log-magnitude spectrum of the CT slice.
    - Apply a **Butterworth Lowpass Filter (BLPF)** or **Homomorphic Filter** to attenuate high-frequency noise and correct illumination gradients across the abdominal cavity.
4. **Validation:** Quantify enhancement using **Peak Signal-to-Noise Ratio (PSNR)** and **Contrast-to-Noise Ratio (CNR)** between soft tissue and surrounding background.

Phase 2: Morphological Processing & Organ Segmentation

Goal: Isolate the target organ (Liver or Kidney) and lesion masks from surrounding anatomical structures.

1. **Morphological Operations:**
    - Design custom Structuring Elements (SE) for Erosion, Dilation, Opening, and Closing to disconnect touching organs (e.g., liver from stomach) and fill internal void spaces in organ masks.
2. **Segmentation Strategy Comparison:** Segmentation then morphological
    - Implement and compare **at least two** classical segmentation approaches:

- **Strategy A:** Multi-Otsu Thresholding or Adaptive Local Thresholding.
- **Strategy B:** Marker-Controlled Watershed Segmentation OR Seeded Region Growing initialized via spatial centroids.

3. **Mask Validation:** Measure overlap precision between predicted masks and expert ground-truth annotations using:
    - **Dice Similarity Coefficient (DSC):**

$$\text{Dice} = \frac{2|A \cap B|}{|A| + |B|}$$

    - **Jaccard Index / Intersection over Union (IoU)**

Phase 3: Feature Extraction & Pattern Recognition

Goal: Extract informative visual features from segmented ROIs and classify tissue pathology.

1. **Feature Extraction:**
    - **Extract the features that you feel are important for object detection. E.g**
    - **Morphological/Shape Descriptors:** Compactness, Circularity, Eccentricity, Area, Perimeter-to-Area ratio, Solidity.
    - **Grey-Level Co-occurrence Matrix (GLCM) Texture:** Energy, Contrast, Homogeneity, Correlation, and Entropy computed across multiple angles ($\theta = 0°, 45°, 90°, 135°$).
    - **First-Order Attenuation Statistics:** Mean HU intensity, Variance, Skewness, Kurtosis within the segmented region.
2. **Pattern Classification:**
    - Formulate normalized feature vectors $\mathbf{x} = [f_1, f_2, \ldots, f_d]^T$.
    - Implement a classical pattern classifier, **DO NOT** use pre-trained or deep learning models.
    - Perform a **70/30 Train/Test split** utilizing 5-fold cross-validation.
3. **Diagnostic Evaluation:** Generate a **Confusion Matrix** and report Classification Accuracy, Sensitivity (Recall), Specificity, and F_1-score.

7.0 DELIVERABLES

In addition, a supporting document is to be produced to reflect on the methodologies and implementation tasks applied in the development of the application.

- In terms of documentation, you are required to describe and justify the theoretical information about the image processing and computer vision algorithms used.
- To fulfil the objective of this coursework, you are also required to describe the results of testing using various test images.
- Pay particular attention in providing critical analysis for cases of images that fail.

7.1 SOURCE CODE:

Fully commented, modular Python (OpenCV/scikit-image/SimpleITK) or a notebook executing reproducibly from raw CT slice ingestion to final classification.

All test files (e.g. *.jpg, *.gif, etc.) are to be provided.

7.2 PROTOTYPE APPLICATION Output: original; mask; original with the mask line (red and green)

A working prototype with the following features as minimum:

1. Load image/group of images
2. Show the segmented mask for each organ.
3. Overlay each detected mask (Red contour) and the ground truth (green contour) on the original image, for each organ separately, and for all organs in another view.
4. Display the diagnostic evaluation metrics in terms of DICE value for segmentation, accuracy, F1, recall for classification.

7.3 DOCUMENTS: PROJECT REPORT

As part of your assessment, you will have to submit a project report which includes the following:

- Table of contents
- **Contribution matrix signed by all members**
- Abstract: The objectives of this project (200-300 words)
- Introduction: Analysis of the problem
- Short literature review for some of the previous work.
- Description and justification of the proposed algorithms/methods/techniques
- Experimental Results
- Description and discussion on the obtained results
- Critical comments, analysis and future work direction
- Conclusion
- List of References (Scholarly references, not wiki, personal websites, blogs, forums)

- Must include mathematical formulations, block diagrams, raw vs. windowed/enhanced visual outputs, segmentation overlay masks, and GLCM feature analysis tables.

❖ The front cover should be presented with the following details:

a) Group members Names and corresponding Student ID numbers
b) Intake code
c) Module Code and Module Name
d) Assignment Title
e) Date Completed (the date the report is due to be handed in)

Note:

❖ The length of the report should be approximately **5000 words.**

❖ You may source information from the Internet. Cite all sources.

❖ All in-text citations and in the References must be made using the APA Convention.

7.4 INDIVIDUAL PRESENTATION & RECORDED VIDEO

❖ Create appropriate slides to explain your approach and contribution in addition to the outcome with example of the output.

❖ Record a **10-15 min** video to show the following:

- You can use your **slides** to explain your solution and contribution.
- **Live demo** for your working prototype to show your algorithms and how it works including the results.
- Open your **source code** in your development environment and explain your approach via the implemented functions and algorithms

8.0 ASSESSMENT CRITERIA

❖ The project will be assessed based on the following key areas:

	Criteria	Marks Allocated
1	Objectives, introduction, literature, Description and justification of the candidate techniques/methods/algorithms	20%
2	Detailed implementation, code quality, and methodology	30%
3	Experimental Results & Discussion	20%
4	Critical analysis	20%
5	Overall project and report	10%
	Assignment Total	**100%**
6	Prototype (Video, Demo, Presentation, and individual contribution)	20%

9.0 DEVELOPMENT TOOLS

❖ Development for the assignment is expected in **Python**. However, alternative programming languages, libraries and tools may be used with justification & discussion with the teacher.

10.0 ACADEMIC INTEGRITY

❖ You are expected to maintain the utmost level of academic integrity during the duration of the course.

❖ **Plagiarism** is a serious offence and will be dealt with according to APU's regulations on plagiarism.

11.0 WORKLOAD MATRIX:

All members are required to sign off your workload matrix, before submitting your assignment. In addition to the workload matrix, a maximum of one page per member (titled with student name and TP number) for individual reflection and detailed contribution and lessons learned.

Note: For presentation, every single member of the group has to do the presentation individually.